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Direct and indirect effects of disturbance on net primary production in giant kelp forests

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Abstract

Disturbances can strongly impact the biomass, abundance, and size and age structure of plant populations, with cascading consequences for net primary production (NPP). Studies investigating sources of interannual variation in annual NPP have largely focused on how interannual fluctuations in resources control plant growth throughout the year, without considering the extent to which interannual fluctuations in disturbance impact the production potential of the system. To address this gap, we evaluated how interannual variation in disturbance affects annual NPP via its three constituent parts, initial production potential, recruitment, and growth in southern California (USA) kelp forests, which are among the world's most productive ecosystems. We estimated mass-specific growth rates and NPP of giant kelp (*Macrocystis pyrifera*) from a model of biomass dynamics developed from monthly field measurements at three sites over a 20-year period. We discovered that initial production potential (i.e., foliar standing biomass at the start of the growing season) and recruitment of new plants accounted for 46% and 26% of the observed variation in annual NPP, respectively. By contrast, the annual growth rate of giant kelp, which averaged 2%–6% day^{-1}, did not explain significant variation in annual NPP despite substantial interannual variation in environmental conditions (e.g., temperature) and nutrient availability during the study period. A piecewise structural equation model explained 82% of the variation in annual NPP and revealed how disturbance from winter waves and sand burial of reef substrate indirectly influenced NPP via their direct and indirect effects on initial production potential and recruitment. Our results highlight the critical but understudied role of disturbance in altering production potential and recruitment to control ecosystem productivity, and the importance of long-term studies in gaining a mechanistic understanding of these processes under intensifying disturbance regimes.

KEYWORDS
biomass, disturbance, giant kelp *Macrocystis pyrifera*, growth, kelp forest, net primary production, recruitment, resources, sand, sea urchin grazing, shading

INTRODUCTION

Net primary production (NPP), the amount of plant matter produced per unit area per unit time, is the foundation for nearly all food webs and ecosystems, and understanding NPP dynamics has long been a fundamental goal in ecology (Leith & Whittaker, 1975). Studies of sources of temporal variation in NPP have largely focused on how resource availability drives plant growth rate (Avolio et al., 2020; Knapp et al., 2017). The role of disturbance on interannual variation in NPP is not well described, despite the potential for disturbance to be a dominant control on NPP (Bernhardt et al., 2022). Resolving how and when disturbance affects variation in NPP is increasingly urgent as humans alter disturbance regimes, and changes in disturbance frequency and severity are forecasted for many regions (Ummenhofer & Meehl, 2017).

Disturbance can influence the constituent components that underlie annual NPP: (1) the initial production potential of the vegetation (i.e., meristem density, foliar area/biomass at the beginning of the growth year, which determines the capacity to generate new growth), and the increase in production during the year due to (2) recruitment and (3) growth (Figure 1a). Disturbance may shift the relative importance of NPP components directly by decreasing production potential or recruitment, or indirectly by changing resource availability and the competitive interactions that influence recruitment and growth. Acute disturbances that directly remove vegetation, such

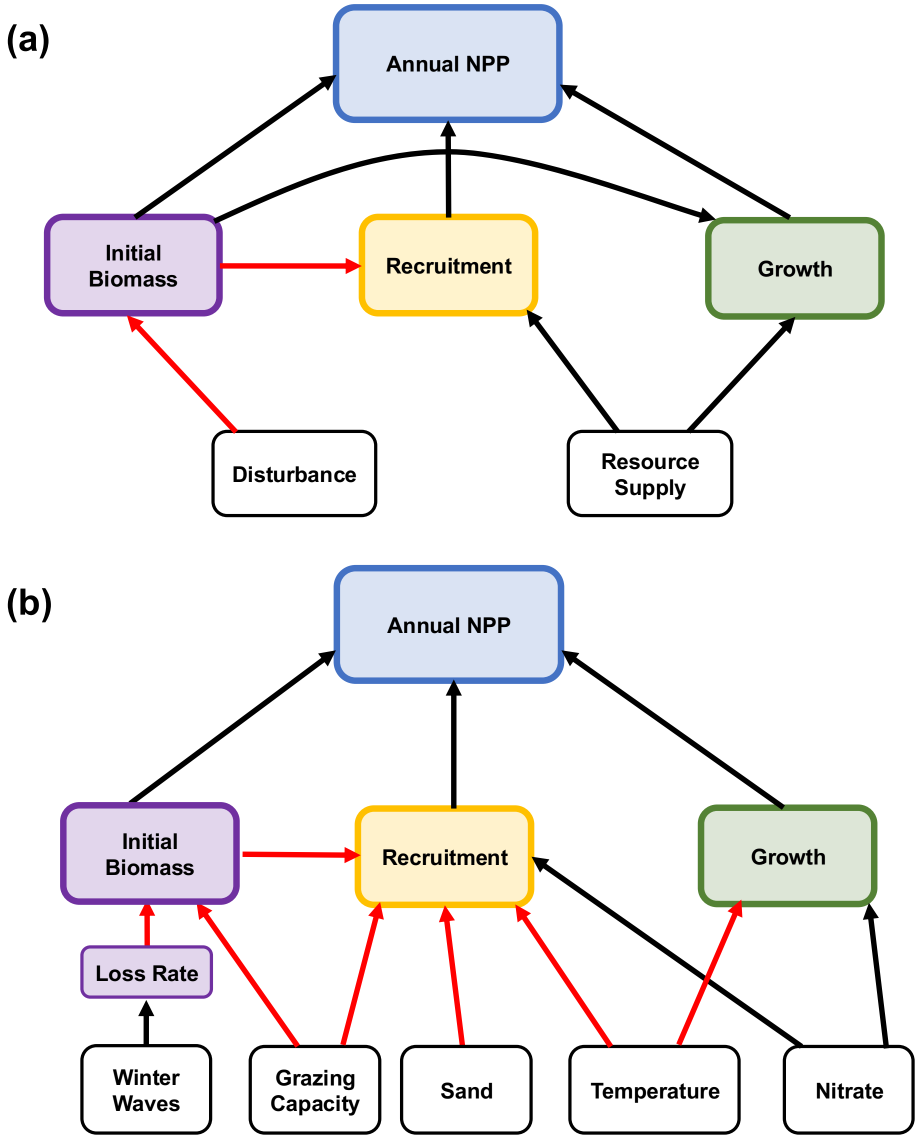

FIGURE 1 Conceptual models showing factors influencing interannual variability in NPP by giant kelp. (a) Hypothesized relationships originally proposed by Reed et al. (2008) and (b) variables and paths used in this study to test the hypothesized relationships in (a). Black arrows indicate mechanisms hypothesized to be positive while red arrows indicate negative paths.

as forest fires (Goulden et al., 2011), landslides (Shiels & Walker, 2013), and ocean waves (Ebeling et al., 1985), can impact annual NPP if not compensated by increases in recruitment and growth later in the growing season. Less intense or more prolonged vegetation disturbances can have more subtle effects, including indirect enhancement of recruitment and growth by reducing resource competition, which can partially or entirely compensate for NPP reductions caused by direct vegetation loss (Reed et al., 2008; Riutta et al., 2018).

Despite progress towards understanding how disturbance affects interannual variation in NPP, studies have been limited by the long durations necessary to observe multiple cycles of disturbances, recruitment, and recovery (Stephenson & van Mantgem, 2005). Much of the empirical research stems from systems characterized by low-lying biomass and limited vertical structure that experience frequent cycles of disturbance and recovery, such as grasslands and streams (Avolio et al., 2020; Bernhardt et al., 2022). Results from such structurally simple systems may not be readily applicable to systems with more complex vertical layering such as forests, which are composed of relatively long-lived trees and relatively long disturbance intervals. Consequently, the primary means for investigating how disturbance affects interannual NPP variation in forests and similar ecosystems has been to use productivity models (Seidl et al., 2011; Sturtevant & Fortin, 2021). The limitations of these methods and the scarcity of empirical data on the pathways by which disturbance alters NPP are pressing challenges as disturbances become more frequent and severe (Gaiser et al., 2020).

Several of the challenges to understanding how disturbance affects NPP in ecosystems with canopy-forming plants can be overcome through study of marine forests formed by large kelps in the order Laminariales, such as the giant kelp *Macrocystis pyrifera* (Reed et al., 2015). Giant kelp extends from the seafloor to the ocean surface and dominates the understory to control ecosystem NPP (Castorani et al., 2021). Giant kelp forests can be as tall as many terrestrial forests (Schiel & Foster, 2015) and on an areal basis are among the most productive ecosystems on Earth (Pessarrodona et al., 2022). However, a key difference between giant kelp and most trees that form terrestrial forests is that giant kelp is a short-lived foundation species that lives only 1–5 years (Dayton et al., 1992). Its short life span is attributed in large part to its size, which increases its susceptibility to being dislodged by large waves during winter storms, to its low tolerance to thermal and nutrient stress during warm, nutrient-poor conditions, and to its vulnerability to episodes of intense grazing by sea urchins (Schiel & Foster, 2015). As a result, most giant kelp stands disappear at least once every 4–6 years (Castorani et al., 2015; Cavanaugh et al., 2011). Relatively frequent recruitment and high growth rates during less stressful conditions enable giant kelp to recover rapidly from disturbance. Such frequent cycles of disturbance and recovery make giant kelp forests particularly well suited for investigating the effects of disturbance on NPP in structurally complex systems. Knowledge gained from such studies is becoming increasingly important as kelp forest loss due to disturbances and rising ocean temperatures has occurred in many regions of the world (Wernberg et al., 2020), and climate change may result in warmer seas and more destructive storms that depress coastal productivity (Ummenhofer & Meehl, 2017).

In our previous work, we used a four-year spatially replicated (three sites) time series of giant kelp biomass, recruitment, and growth to develop a conceptual model of the factors controlling giant kelp NPP (Reed et al., 2008; Figure 1a). Here, we test this conceptual model with an additional 16 years of time series data that encompassed multiple disturbances and broad fluctuations in environmental conditions. We reevaluated our prior hypotheses and conclusions; determined the importance of variation in abiotic conditions, resource availability, and disturbance; and quantitatively partitioned the direct and indirect pathways controlling interannual variability in giant kelp NPP. Specifically, we evaluated the relative importance of different forms of disturbance and resource availability in altering initial production potential, recruitment, and growth as drivers of variation in annual NPP (Figure 1b). We predicted that winter wave disturbance would have direct and indirect effects on NPP, directly by increasing giant kelp loss in winter and decreasing the standing biomass at the beginning of the growing season (i.e., initial production potential), and indirectly by enhancing recruitment via increases in light due to a reduction in canopy shading. We also predicted that the positive influence of recruitment on NPP would be impacted by variation in other factors, expecting a positive relationship with the amount of suitable substrate and negative effects from grazing sea urchins and from prolonged periods of warm, nutrient-poor water. To test our predictions, we analyzed measurements of initial biomass (i.e., production potential), growth, recruitment, and NPP of giant kelp collected monthly in fixed plots at three sites over 20 years. We also measured environmental covariates (e.g., wave height, nutrients, temperature, substrate availability) thought to influence giant kelp biomass, recruitment, and growth. In this way, our study uniquely quantified the direct and indirect effects of variation in resource availability, temperature, and disturbance on plant biomass dynamics and vital rates to understand the controls on interannual variation in NPP over multiple generations spanning two decades.

METHODS

Study system

Giant kelp is the world's largest and most widely distributed kelp species, occurring on shallow temperate rocky reefs in the northern and southern hemispheres (Schiel & Foster, 2015). Mature giant kelp sporophytes (hereafter plants) have a holdfast that anchors the plant to the seafloor and buoyant fronds that can create a dense canopy at the sea surface. Since 2002, divers with the Santa Barbara Coastal Long Term Ecological Research project (SBC LTER) have surveyed giant kelp monthly at three sites in southern California, USA: Arroyo Quemada (34°28′08″ N, 120°07′17″ W), Mohawk (34°23′40″ N, 119°43′48″ W), and Arroyo Burro (34°24′00″ N, 119°44′40″ W). Surveys include nondestructive measurements of giant kelp density, size, biomass, recruitment, and plant and frond loss rates in permanent plots at 5–10 m depth (plot area = 200 m^2 at Arroyo Quemada and Mohawk; 480 m^2 at Arroyo Burro; Rassweiler et al., 2024).

Factors affecting giant kelp biomass at the beginning of the growth year

We chose April as the beginning of the kelp growth year when giant kelp biomass in our region is usually near its annual minimum due to winter wave disturbance (Reed et al., 2008). In each plot, all plants were measured and dry mass estimated using allometric relationships (Rassweiler et al., 2024). We calculated biomass loss rates for 10–15 tagged individuals per site during each monthly sampling interval in winter (December–March) to examine the relationship between initial biomass and the rate of biomass loss during the preceding winter (Rassweiler et al., 2024). We tested how winter wave disturbance affected winter loss rates using the average maximum daily significant wave height (H_s = mean height of the largest one-third of waves) during the winter at each site, which we estimated using an empirical wave model (Bell, 2024).

Biological disturbance such as grazing also has the potential to reduce the initial biomass and recruitment of giant kelp. Sea urchins are the most important grazers in kelp forests worldwide, and the effects of their intensive grazing on giant kelp are well documented (Schiel & Foster, 2015). Measuring actual grazing rates of sea urchins within our large plots over the course of our 20-year time series was not feasible. Moreover, estimating the grazing intensity of sea urchins based on their density or biomass is complicated by the fact that urchins display two distinct foraging modes: (1) active grazing on living kelp and (2) passive foraging on detrital drift kelp (Harrold & Reed, 1985). As a result, the effects of sea urchin grazing on giant kelp are most evident when their consumption exceeds the supply of kelp detritus (Schiel & Foster, 2015). Therefore, we evaluated the capacity of sea urchins to consume giant kelp (i.e., "grazing capacity" sensu Rennick et al., 2022) as a source of interannual variation in giant kelp NPP rather than their actual grazing intensity, recognizing that the relationship between grazing capacity and grazing intensity varies with the availability of detrital kelp. We calculated the grazing capacity of the two common sea urchin species (*Mesocentrotus franciscanus* and *Strongylocentrotus purpuratus*) at our sites as the product of their biomass and mass-specific grazing rates of giant kelp. We estimated the wet biomass of red and purple sea urchins by applying species-specific size–mass relationships to annual measurements of their density and size collected in situ at each site in summer (Reed & Miller, 2024a). We used mass-specific grazing rates measured by Rennick et al. (2022) in the laboratory for each sea urchin species (9.1 and 20 mg of giant kelp per g of sea urchin per m^2 per day for red and purple urchins, respectively).

Factors affecting giant kelp recruitment

Giant kelp recruits first appear as small (<5 cm), single blades in the spring (April and May) at the start of the growing season (Dayton et al., 1992; Ebeling et al., 1985; Reed & Foster, 1984). They rapidly grow to >1 m tall by summer when they are first captured by our sampling. Thus, we estimated recruitment as the mean density of plants measuring 1–2 m tall with ≤2 fronds in summer (June, July, and August), which is when plants in this size class are most abundant at our study sites (Reed et al., 2008).

Light near the seafloor is critically important for giant kelp recruitment, and shading by adult giant kelp is widely known to inhibit the recruitment of juvenile plants (reviewed in Schiel & Foster, 2015). Because the amount of light near the seafloor is inversely related to giant kelp biomass at our study sites (Castorani et al., 2021), we used initial kelp biomass as a proxy for light availability in April when small recruits typically first appear (Harris et al., 1984).

Successful recruitment of giant kelp also depends on hard substrate for attachment and cool, nutrient-replete water which supports juvenile growth (Castorani et al., 2021; Dean & Jacobsen, 1984). The amount of hard substrate at our study sites can vary substantially due to sand movement, while ocean temperature and nitrate vary seasonally and interannually with oceanographic conditions. We used a uniform-point-contact method to estimate the percent cover of sand at each site during the summer recruitment period (Reed & Miller, 2024b). Using bottom water temperature at each site measured every 10 min with data loggers (Stowaway Onset TidbiT, accuracy ±0.2°C; Onset Computer, Massachusetts, USA; Reed & Miller, 2024c), we calculated the number of days during the recruitment period when the mean water temperature was ≥19°C, above which growth of juvenile giant kelp in southern California is inhibited (Dean & Jacobsen, 1984). We estimated the concentration of dissolved inorganic nitrogen (DIN = nitrate + nitrite) from bottom temperature using relationships developed at our sites with data from bottom-moored temperature sensors and in situ DIN auto-analyzers (Arroyo Quemada: McPhee-Shaw et al., 2007; Mohawk and Arroyo Burro: Fram et al., 2008). The number of days in which mean DIN concentrations (hereafter "nitrate") were <1 μmol L^{-1} from April through August was used to investigate the relationship between giant kelp recruit density and nutrient availability. Nitrate concentrations <1 μmol L^{-1} suppress giant kelp growth in southern California (Gerard, 1982).

Factors affecting giant kelp growth

We estimated the mean daily growth rate of giant kelp using monthly field measurements of standing biomass and biomass loss rates at each site in a model of biomass dynamics developed by Rassweiler et al. (2024). Daily mass-specific growth rates were averaged over all days in a growth year (April 1 in year t to March 31 of year t+1) to estimate an annual average growth rate (day^{-1}) for each site.

To examine the extent to which annual growth of giant kelp varied with temperature and nitrate concentrations, we again used the number of days with a mean daily temperature ≥19°C, and a mean nitrate concentration <1 μmol L^{-1} as response variables. Three site–year combinations with <357 days of temperature data were dropped from both the recruitment and growth analyses ($n = 57$, equivalent to 95% of all site–year observations).

As in our prior study (Reed et al., 2008), we determined the extent to which initial biomass, recruitment, and growth explained interannual variation in giant kelp NPP (Figure 1a). Daily estimates of NPP were summed over all days in a growth year to produce an estimate of annual NPP (in kilograms of dry mass per square meter per year) for each site from 2003 through 2023 (with the exception of 2020 due to limitations on research imposed by the COVID-19 pandemic).

Statistical analyses

All statistical analyses were performed using R 4.4.2 (R Core Team 2024). We used linear models (LMs) and generalized linear models (GLMs) to evaluate the strength of the directional relationships depicted in Figure 1b. Specifically, we fit five models quantifying the following relationships: (1) the loss rate of giant kelp biomass during winter as a function of winter wave height; (2) initial giant kelp biomass as a function of urchin grazing rate and the winter loss rate; (3) growth as a function of temperature stress, nitrate stress, plant density, and plant size; (4) annual NPP as a function of initial giant kelp biomass, recruitment, and growth, and (5) recruitment as a function of sand cover, urchin grazing rate, initial biomass, thermal stress, and nitrate availability. All models included a fixed effect of site (three levels) to account for unmeasured among-site variation (Byrnes & Dee, 2025).

All models were LMs except the recruitment model (model 3), which required a negative binomial GLM due to non-normal residuals. The annual recruitment data were based on count observations but were non-integer values due to averaging across sampling periods (mean density of recruits per square meter). To conform with the assumptions of the negative binomial distribution, we multiplied by 1000 and rounded to the nearest integer; results were robust to different choices for this scaling constant. We back-transformed recruitment model predictions for data visualization. We evaluated collinearity of predictors using the generalized variance inflation factor (GVIF; Fox & Monette, 1992) using the *car* package. All GVIF values were <2.52, indicating minimal collinearity of predictor variables.

We assessed the predicted distribution of outcomes versus the observed distribution of response variables in addition to ensuring all models satisfied assumptions regarding collinearity: linearity, homoscedasticity, and normality of residuals and influential observations for LMs using the performance package (Lüdecke et al., 2021). For the GLM, we ensured satisfaction of these assumptions using simulated quantile residuals with the *DHARMa* package (Hartig, 2024). We examined the temporal autocorrelation of the LM/GLM residuals by applying the Ljung–Box test. In all cases, residuals were uncorrelated over time (Ljung–Box $p > 0.05$).

To evaluate the direct and indirect drivers of annual NPP, we built a piecewise structural equation model

(SEM) using the *piecewiseSEM* package (Lefcheck, 2016; Figure 1b). This model included the relationships specified above for giant kelp loss rate, initial biomass, annual recruitment, and NPP (Figure 1b). We used Fisher's C to determine whether the hypothesized model structures did not exclude any conditionally dependent (i.e., relevant) paths (Lefcheck, 2016; Shipley, 2000). Because site–year combinations used in the SEM require data for every variable, site years without kelp were omitted from this analysis.

To compare the strength of associations corresponding to different hypotheses, we used standardized coefficients corresponding to Pearson correlation coefficients (Grace & Bollen, 2005) for LMs or calculated according to *effectsize* package for the GLM paths. We also calculated partial R^2 to explain the proportion of variance explained by each predictor variable in each model using *sensemakr* for LMs and *rsq* for GLMs. R Code (Beckley, 2026) is available in Zenodo at https://doi.org/10.5281/zenodo.18190569.

RESULTS

Factors affecting giant kelp biomass at the beginning of the growth year

Over the 20-year time series, giant kelp biomass at the start of the growing season varied greatly (Figure 2a), with the standard deviation exceeding the mean at all three sites (i.e., coefficient of variation = 1.54, 1.33, and 1.18 at Arroyo Burro, Arroyo Quemada, and Mohawk, respectively). As predicted, winter wave disturbance was positively associated with the loss of giant kelp biomass during winter (+0.27; standardized coefficient; Table 1a), but it only accounted for 8% of the interannual variation in the average daily loss rate during winter (Appendix S1: Figure S1a). The negative relationship between initial kelp biomass and winter biomass loss rates in turn accounted for 25% of the interannual variation in initial kelp biomass (Table 1b; Appendix S1: Figure S1a). Initial kelp biomass was unrelated to urchin grazing capacity (Table 1b; Appendix S1: Figure S2).

Factors affecting giant kelp recruitment

Giant kelp recruitment varied substantially among sites over time, ranging from 0 to 1.4 plants m^{-2} (Figure 2b). GLM results indicated annual giant kelp recruitment was strongly influenced by resource availability. Consistent with our predictions, density of giant kelp recruits was inversely related to the amount of giant kelp biomass

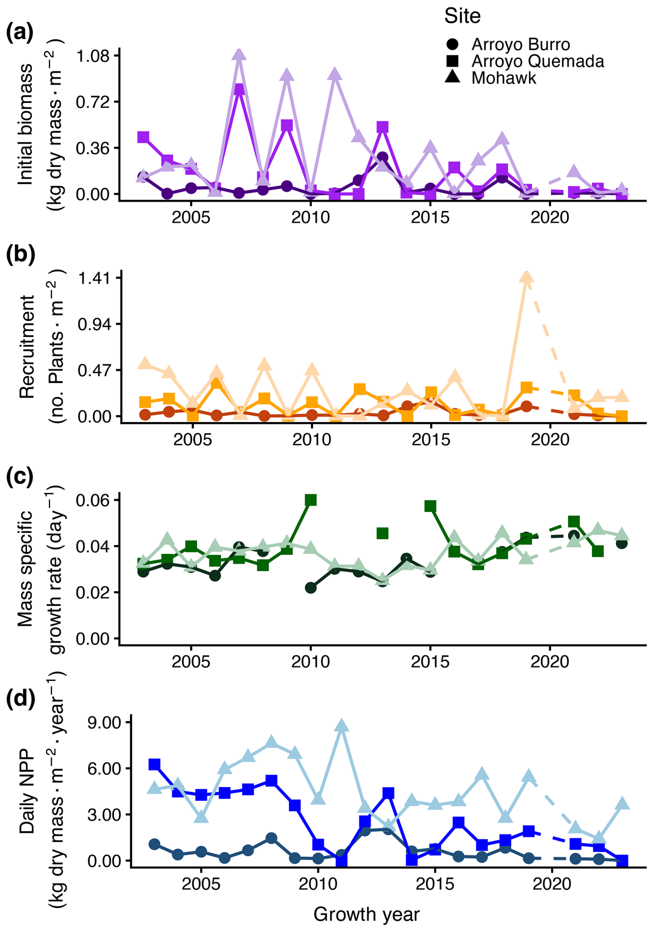

FIGURE 2 Annual values at the three study sites for giant kelp: (a) biomass at the beginning of the growth year, (b) average recruitment density during the year, (c) average mass-specific growth rate during the year, and (d) annual NPP. Dashed lines represent years with missing data.

present at the beginning of the growth year, suggesting that intraspecific shading by plants surviving winter disturbance inhibited recruitment (Table 1c; Appendix S1: Figure S1b). The negative relationship between giant kelp recruit density and sand cover (−1.03) was stronger than the negative relationship between recruit density and initial giant kelp biomass (−0.91). However, sand cover accounted for only 6% of the initial variation in giant kelp recruitment compared to 14% for initial giant kelp biomass (Table 1c; Appendix S1: Figure S1b). Much like initial giant kelp biomass, giant kelp recruitment was unrelated to the grazing capacity of sea urchins (Table 1c; Appendix S1: Figure S3a). The lack of this relationship was unrelated to the percent cover of sand, as sand cover varied greatly when urchin grazing capacity was low, and urchin grazing capacity varied greatly when sand cover was low (Appendix S1: Figure S4). Similarly, we did not detect any influence of nitrate stress or warm water on recruitment, despite extended periods of

TABLE 1 Results from linear (LM) and generalized linear (GLM) regression models.

Response variable and source of variation	df	F	χ^2	p	Partial R^2	Standardized coefficient
(a) Instantaneous loss rate						
Large waves	1	4.2		**0.04**	0.08	0.27
Site	2	1.2		0.32		
Residual	52					
(b) Initial biomass						
Instantaneous loss rate	1	16.7		**<0.01**	0.25	−0.47
Urchin grazing capacity	1	2.9		0.09	0.05	0.25
Site	2	2.5		0.09		
Residual	51					
(c) Kelp recruitment						
Sand cover	1		6.9	**<0.01**	0.06	−1.03
Urchin grazing capacity	1		0.05	0.83	0.00	−0.05
Temperature	1		1.03	0.31	0.05	−0.22
Nitrate	1		1.5	0.22	0.04	0.37
Initial biomass	1		18.9	**<0.01**	0.14	−0.91
Site	2		2.3	0.31	0.10	
Residual	49					
(d) Kelp growth rate						
Kelp plant density	1	1.02		0.32	0.02	−0.18
Kelp plant size	1	7.5		**<0.01**	0.15	−0.39
Nitrate	1	0.02		0.89	0.0	−0.03
Temperature	1	0.02		0.89	0.0	0.02
Site	2	2.0		0.15		
Residual	42					
(e) Kelp net primary production						
Initial kelp biomass	1	38.4		**<0.01**	0.46	0.68
Annual kelp recruitment	1	15.8		**<0.01**	0.26	0.48
Annual kelp growth	1	2.7		0.1	0.06	−0.16
Site	2	7.4		**<0.01**		
Residual	46					

Note: $p < 0.05$ should are bolded to highlight a statistically significant result. Test statistics are reported as F values for LMs and χ^2 values for GLMs. Kelp recruitment (model c) was the only response modeled using a GLM.

warm nutrient-poor conditions in several years (Appendix S1: Figure S3b,c).

Factors affecting giant kelp growth

Relatively high rates of growth ranging from ~2% to 6% day^{-1} were observed at all three sites throughout the 20-year time series (Figure 2c). As expected, giant kelp growth rate depended on plant size, as smaller plants tended to grow relatively faster than larger plants; average plant size accounted for 15% of the interannual variation in giant kelp growth rate (Table 1d; Appendix S1: Figure S1c). However, contrary to our predictions, growth was unrelated to plant density or to thermal and nitrate stress (as indicated by the number of days above the temperature threshold and below the nitrate threshold known to support growth; Table 1d; Appendix S1: Figure S5).

Drivers of interannual variation in giant kelp NPP

Giant kelp NPP was significantly and positively related to initial giant kelp biomass and giant kelp recruitment

(Table 1e; Appendix S1: Figure S6a,b). The relationship between NPP and initial giant kelp biomass was stronger than the relationship between NPP and giant kelp recruitment (+0.68 vs. +0.48), accounting for 46% and 26% of the interannual variation in NPP, respectively. Annual growth rate remained relatively high throughout the study and did not explain any significant variation in annual NPP (Table 1e; Appendix S1: Figure S6c). Consequently, predictors of giant kelp growth were not explored in the SEM.

The SEM analysis explained 82% of the variation in annual NPP (Figure 3, Appendix S1: Table S1). Initial giant kelp biomass influenced NPP through both direct and indirect pathways. The direct effect was strongly positive (+0.75), indicating that greater initial biomass generally leads to higher NPP, and was greater than the direct positive effect of recruitment (+0.42). However, a negative indirect effect of initial biomass on annual NPP via its strong negative effect on recruitment (−0.81) reduced the overall positive effects of initial biomass on annual NPP to +0.41.

The SEM also revealed that disturbance was a major driver of annual NPP through its direct and indirect effects on initial giant kelp biomass and annual recruitment. The direct and indirect effects of wave disturbance on initial biomass accounted for 45% of the overall variation in initial biomass, while the direct and indirect effects of wave disturbance, sand cover, and nitrate accounted for 89% of the annual variation in recruitment. Unlike the GLM which showed no direct relationship between nitrate stress and recruitment (Table 1c; Appendix S1: Figure S3c), the SEM detected a marginally significant positive path from nitrate stress to recruitment (Appendix S1: Table S1, Figure 3). The SEM also identified three missing paths: (1) between giant kelp loss rate and annual recruitment, (2) between winter waves and initial biomass, and (3) between winter waves and annual NPP (Fisher's $C = 29.07$, $p = 0.514$, df $= 30$). Thus, some processes may not be fully accounted for in our data and study design. For example, our measurement of the loss rate may not capture the true loss of giant kelp biomass or other effects on NPP by winter wave disturbance.

DISCUSSION

Our long-term study demonstrated that disturbance-driven changes in giant kelp NPP were primarily

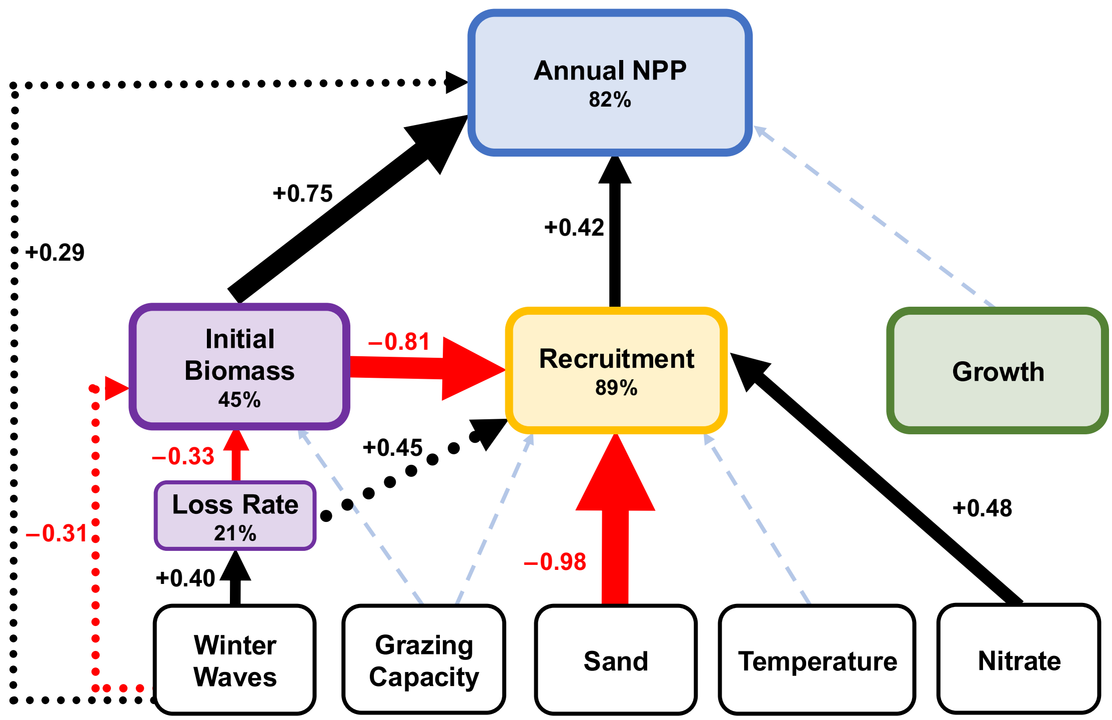

FIGURE 3 Results of structural equation model (SEM) showing the direct and indirect effects of predictors on interannual variability in NPP. Values (percent) within boxes refer to the coefficient of determination (R^2) of individual models. Black arrows with positive regression coefficients indicate positive paths and red arrows with negative regression coefficients indicate negative paths. Solid black or red arrows represent significant hypothesized paths, while dotted red or black arrows represent unplanned paths that were added based on tests of directed separation. The widths of the arrows scale with the magnitude of the standardized regression coefficients, which represents the effect size of that predictor. Nonsignificant paths are shown using blue dashed lines. Added paths based on tests of directed separation include: (1) winter waves to biomass, (2) kelp loss rate to recruitment, and (3) winter waves to NPP. $\chi^2 = 13.669$, df $= 15$, $p = 0.551$.

mediated through changes in production potential measured as the foliar biomass at the start of the growth year. By collecting detailed monthly plant measurements and tracking biological and environmental drivers at multiple sites over two decades, we found that the amount of foliar biomass at the beginning of the growth year varied greatly among years and explained much more of the interannual variation in giant kelp NPP than factors controlling the rate of biomass increase during the year via recruitment and growth. Wave disturbance that increased the loss of biomass in winter reduced initial biomass (i.e., production potential at the beginning of the growing season), and that loss of biomass led directly to a reduction in annual NPP. However, that same loss of biomass increased annual NPP indirectly, albeit by a smaller amount, by enhancing recruitment. In this case, the removal of standing biomass by waves decreased competition for light and space, which increased recruitment success thereby enhancing NPP. Such direct and indirect effects of disturbance on the production potential of vegetation are usually not considered when investigating sources of interannual variation in NPP (Knapp & Smith, 2001; Mohamed et al., 2004). In these less frequently disturbed systems, variation in annual NPP is most often explained by fluctuations in environmental conditions and resource availability that affect plant growth, such as precipitation, temperature, and nutrients (Craine et al., 2012; LeBauer & Treseder, 2008; Zeng et al. 2022).

Severe disturbances that reduce the production potential of a system by removing plant biomass, such as fires, high winds, and large waves (Dayton et al., 1992; Reich et al., 2001; Roberts, 2004), likely have negative effects on ecosystem productivity. Indeed, annual NPP of giant kelp was strongly inversely related to the removal of its biomass by winter waves prior to the start of the growing season. In contrast, we found little evidence that biological disturbance from sea urchin grazing contributed to variation in annual NPP of giant kelp at our sites. Rennick et al. (2022) found a 50-fold reduction in giant kelp biomass when grazing capacity exceeded the production of kelp detritus, which they measured as the biomass of giant kelp lost as fronds and blades per day per unit area. That this condition rarely occurred at our sites (Appendix S1: Figure S7) likely explains why sea urchins were not a significant source of variation in the initial biomass and recruitment of giant kelp in our study.

Our findings on the overriding importance of disturbance in controlling variation in NPP parallel production patterns in rivers and streams dominated by short-lived algae and macrophytes that display rapid growth and undergo frequent cycles of disturbance and recovery due to high flow events that disturb the riverbed (Bernhardt et al., 2022; Blaszczak et al., 2019; Dodds et al., 1996). Systems with longer lived species (e.g., terrestrial forests) display similar patterns of decreased NPP following severe disturbances such as fires (Peters et al. 2013) or intensive logging (Riutta et al., 2018) that remove entire stands. However, disturbance is often not the predominate predictor of variation in annual NPP of these slower growing systems because large stand-removing events occur infrequently (Foster et al., 1998), and recovery times typically exceed a decade (Goulden et al., 2011). Further, the implications of biomass loss from disturbances are nuanced in systems with complex vegetation structure, as effects that remove biomass can increase other vital rates and processes that contribute to NPP (e.g., recruitment and growth; Riutta et al., 2018). For example, Stuart-Haëntjens et al. (2015) discovered that understory NPP compensated for upper canopy production losses following disturbances that removed <60% of the total basal tree area. Similar processes occur in giant kelp forests as giant kelp removal has been shown to result in compensatory increases in NPP by phytoplankton and understory macroalgae (Castorani et al., 2021; Miller et al., 2011).

Our finding of the strong influence of initial giant kelp biomass on NPP has potential implications for management and conservation in light of the growing interest in actively restoring disturbed kelp forests for purposes of recovering their ecological functions. The frequent loss of giant kelp biomass in winter due to wave disturbance has been well documented (Bell et al., 2015; Dayton et al., 1992; Dayton & Tegner, 1984), and differences in year-to-year variation in wave disturbance have been linked to regional differences in giant kelp NPP (Reed et al., 2011). Consequently, restoration approaches focused directly on manipulating population size (e.g., by seeding or outplanting juveniles) are unlikely to have persistent effects, especially in regions where severe wave disturbance consistently overwhelms bottom-up and top-down forcing to control NPP.

The importance of wave disturbance in controlling interannual variation in NPP may not apply universally to canopy-forming kelps. For example, much like giant kelp, bull kelp (*Nereocystis luetkeana*) and dragon kelp (*Eualaria fistulosa*) extend throughout the water column to form a dense canopy at the sea surface. However, most individuals of these species do not survive through winter due to their annual/biennial life history (Konar, 2000; Springer et al., 2010). Consequently, they likely display relatively little interannual variation in biomass at the beginning of the growth year. The same may be true of longer lived kelp species with shorter stature that extend only 1–2 m off the bottom and show a relatively high capacity to withstand dislodgement by large waves (Dayton & Tegner, 1984).

Because canopy-forming species strongly influence resources available on the forest floor, canopy loss can indirectly influence ecosystem productivity by altering the supply of resources that fuel recruitment and growth (Haber et al., 2020; Palik et al., 1997). This is common in kelp forests, where the amount of light reaching the seafloor declines exponentially with kelp biomass (Castorani et al., 2021; Toohey & Kendrick, 2008). Hence, our finding of an indirect positive effect of winter waves on recruitment results from the negative effect of waves on the standing biomass at the beginning of the growing season, which has been shown to inhibit recruitment via shading and preemption of space (Reed & Foster, 1984). The strong negative path between recruitment and sand cover reflects the fact that the survival of kelp recruits is critically dependent on the availability of hard substrate for attachment. Sand cover on shallow reefs can fluctuate greatly (Ebeling et al., 1985), and it varied substantially among sites and years during our study. Collectively, these findings are consistent with our hypothesis that disturbance mediates recruitment by indirectly controlling the availability of critical resources.

Our finding that variation in the annual growth rate of giant kelp failed to explain variation in its annual NPP matches prior analyses of the first 4 years of our time series that used a less comprehensive model of growth and NPP (Reed et al., 2008). This consistency is notable given that the last 16 years in our time series included environmental conditions well outside those observed during the first 4 years, including multiple El Niños and a 2-year warming event that was the most severe on record (Di Lorenzo & Mantua, 2016). Anomalies in nitrate concentrations were negative in every month at our study sites during this warming event, and nitrate concentrations were <1 μmol L^{-1} for 54% of the days compared to 7% of the days in other years (Reed et al., 2016). Nitrate is considered to be the primary nutrient limiting the growth of giant kelp (Zimmerman & Kremer, 1986) and, in the absence of internal reserves and other sources of nitrogen, nitrate concentrations <1 μmol L^{-1} are unable to sustain the growth rate needed to maintain giant kelp populations in southern California (Gerard, 1982). Our finding that annual growth rates of giant kelp were unrelated to the number of warm, nitrate-deplete days suggests that the annual growth at our study sites was not limited by this form of nitrogen. Surprisingly, annual growth rates were relatively steady in comparison to changes in giant kelp initial biomass, even during years with high seawater temperatures and low nitrate concentrations.

Although the growth rate per se (which averaged nearly 4% day^{-1} over the 20-year study) was not a significant predictor of interannual variation in giant kelp NPP, it was nonetheless responsible for the high rates of annual NPP that we observed, which averaged 2.54 kg dry m^{-2} $year^{-1}$ at our sites over the 20-year period. This annual average is approximately five times higher than that measured for giant kelp at the northern poleward edge of its range using a method similar to ours (Bell & Kroeker, 2022) and is comparable to that of other types of kelp forests and temperate terrestrial forests (Pessarrodona et al., 2022), supporting previous assertions that kelp forests are among the most productive ecosystems in the world (Reed and Brzezinski 2009). Unlike terrestrial forests whose productivity is generated by a large and relatively long-lived standing biomass (e.g., 500–600 kg dry m^{-2}; Parresol, 1999), the high production of giant kelp resulted from rapid growth of a relatively small standing biomass (usually <1 kg dry m^{-2}) that turned over about 12 times a year (Rassweiler et al., 2018). Giant kelp NPP at our sites remained relatively high even during anomalously warm, nutrient-poor years because kelp growth rates were largely resilient to such "unfavorable" oceanic conditions. This finding contrasts with results from other, mostly terrestrial systems, in which variation in resource and environmental conditions that affect growth rates drive interannual variation in NPP (Gao et al., 2019; Knapp & Smith, 2001). The primary influence of disturbance-driven changes to production potential on NPP that we observed in giant kelp forests may become more common as the frequency of severe disturbances increases.

Our study highlights the value of kelp forests as a model system for understanding how disturbance shapes temporal and spatial variation in NPP. Productive macroalgae that provide the complex vertical structure of kelp forests usually exhibit rapid recovery following disturbance via high rates of recruitment and growth (Castorani et al., 2015; Dayton et al., 1992). These characteristics make marine forests amenable to studying multiple cycles of disturbance and recovery, unlike their terrestrial counterparts that are often characterized by slower growing species with longer generation times and lengthy recovery intervals. The relatively fine temporal resolution (monthly) and long duration (20 years) of our measurements of biomass change, biomass loss rates, and demography in giant kelp are unique, but necessary, for characterizing many benthic marine systems. And yet, despite frequent disturbance and rapid recovery, it took over a decade to begin to capture the influence of multiple large environmental fluctuations (e.g., large winter waves, sand burial, or prolonged periods of elevated seawater temperature and low nitrate) that contribute to variation in giant kelp NPP. Short-term studies (e.g., <1–3 years), which are the norm for characterizing rocky reef primary productivity (Littler & Arnold, 1980; Miller et al., 2011), typically do not capture such a wide range of environmental conditions, limiting our ability to explore how different factors impact demography and growth to influence NPP.

In conclusion, our results demonstrate how disturbances to canopy-forming vegetation influence productivity dynamics through both direct and indirect pathways. While the loss of biomass due to disturbance can reduce NPP directly, recruitment responses to changes in biomass may either accelerate or delay recovery, depending on the resources available. Understanding NPP dynamics in a frequently disturbed system may become increasingly important and more widely applicable to conservation and restoration as biomass-removing disturbances increase in frequency and severity due to changes in land use and climate.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. MCNC is an editor for *Ecology* and took no part in the editorial review process.

DATA AVAILABILITY STATEMENT

Data are available in the Environmental Data Initiative EDI Data Portal at https://doi.org/10.6073/pasta/b09ac62414a439cab3da207526b8dacf (Coastal Data Information Program and Bell, 2024), https://doi.org/10.6073/pasta/4f4f661ee16e6bb867e58ce82c5a0776 (Rassweiler et al., 2024), https://doi.org/10.6073/pasta/6587ad06e299e566e2092d1268dc206b (Reed & Miller, 2024a), https://doi.org/10.6073/pasta/869317d54d5f7859add452722319d72c (Reed & Miller, 2024b), and https://doi.org/10.6073/pasta/6dfd2d21eb231f83df2abbd6b556acbf (Reed & Miller, 2024c). Code (Beckley, 2026) is available in Zenodo at https://doi.org/10.5281/zenodo.18190569.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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